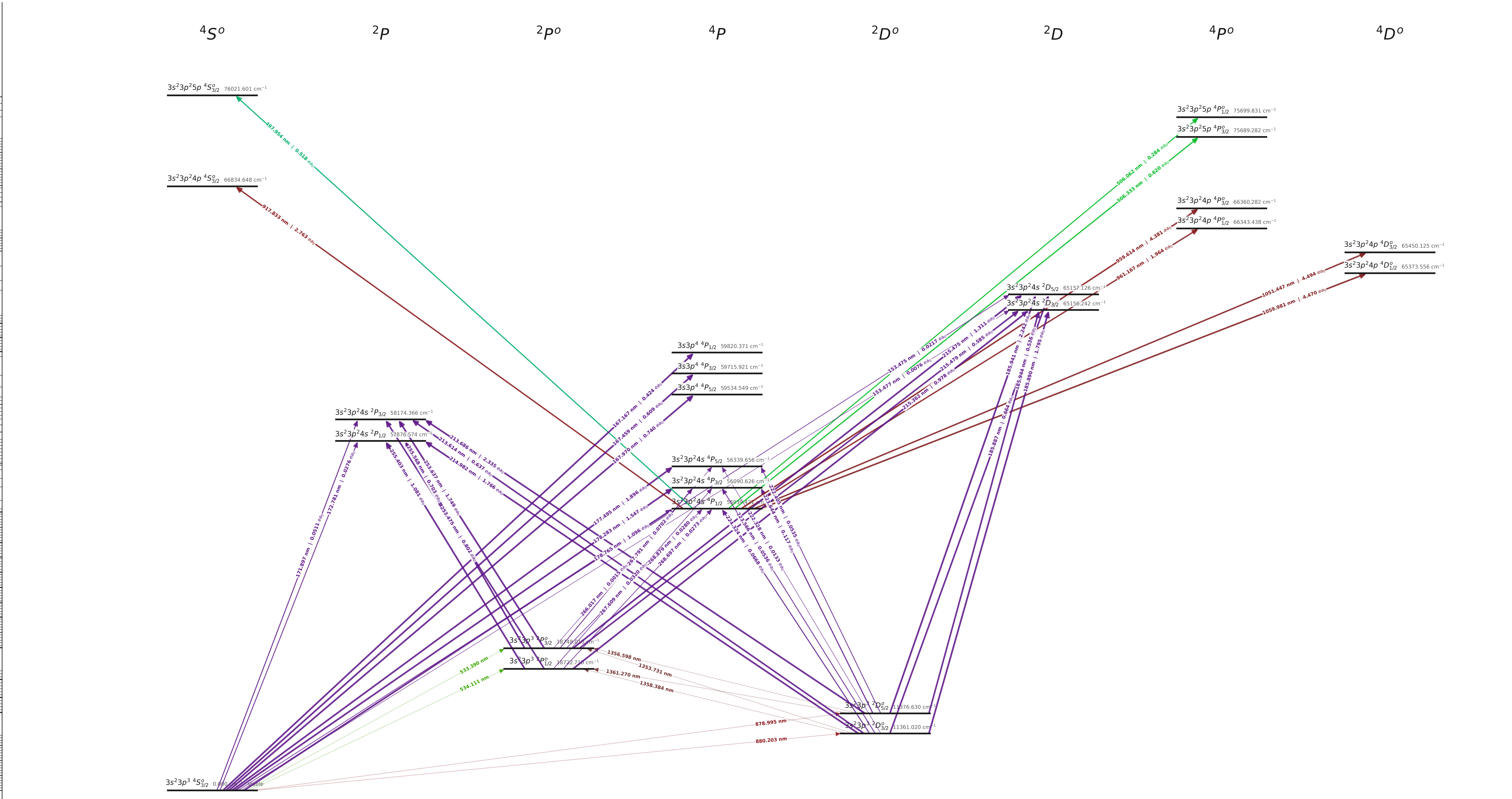


Phosphorus

P (P I). Strongest classified transitions below 2 μm (NIST): 23 levels, 50 lines.

P+ ionisation limit 84580.83 cm⁻¹ = 10.48669 eV (118.230 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	(U er J) (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
3s ² 3p ³ 4S° _{3/2} – 3s3p ⁴ 4P _{1/2}	167.1671	–	1793.3696	0.424	3.900e+07	–	–	
3s ² 3p ³ 4S° _{3/2} – 3s3p ⁴ 4P _{3/2}	167.4595	–	1790.2383	0.609	4.000e+07	–	–	
3s ² 3p ³ 4S° _{3/2} – 3s3p ⁴ 4P _{5/2}	167.9697	–	1784.8009	0.740	3.900e+07	–	–	
3s ² 3p ³ 4S° _{3/2} – 3s ² 3p ² 4s 4P _{5/2}	177.4949	–	1689.0204	1.896	2.170e+08	–	–	
3s ² 3p ³ 4S° _{3/2} – 3s ² 3p ² 4s 4P _{3/2}	178.2829	–	1681.5547	1.547	2.140e+08	–	–	
3s ² 3p ³ 4S° _{3/2} – 3s ² 3p ² 4s 4P _{1/2}	178.7648	–	1677.0216	1.096	2.130e+08	–	–	
3s ² 3p ³ 2D° _{3/2} – 3s ² 3p ² 4s 2D _{5/2}	185.8871	–	1612.7667	0.664	2.320e+07	–	–	
3s ² 3p ³ 2D° _{3/2} – 3s ² 3p ² 4s 2D _{3/2}	185.8901	–	1612.7402	1.795	2.540e+08	–	–	
3s ² 3p ³ 2D° _{3/2} – 3s ² 3p ² 4s 2D _{1/2}	185.9410	–	1612.2987	2.242	2.640e+08	–	–	
3s ² 3p ³ 2D° _{3/2} – 3s ² 3p ² 4s 2P _{3/2}	213.6856	213.6182	1402.9608	2.335	2.830e+08	–	–	
3s ² 3p ³ 2D° _{3/2} – 3s ² 3p ² 4s 2P _{1/2}	214.9819	214.9142	1394.5012	1.766	3.180e+08	–	–	
3s ² 3p ³ 2P° _{1/2} – 3s ² 3p ² 4s 2D _{3/2}	215.3616	215.2939	1392.0423	0.978	4.850e+07	–	–	
3s ² 3p ³ 2P° _{3/2} – 3s ² 3p ² 4s 2D _{5/2}	215.4749	215.4072	1391.3103	1.311	5.800e+07	–	–	
3s ² 3p ³ 2P° _{3/2} – 3s ² 3p ² 4s 2P _{3/2}	253.6374	253.5612	1181.9724	1.749	9.500e+07	–	–	
3s ² 3p ³ 2P° _{1/2} – 3s ² 3p ² 4s 2P _{1/2}	255.4026	255.3260	1173.8033	1.081	7.100e+07	–	–	
3s ² 3p ³ 2P° _{3/2} – 3s ² 3p ² 4s 2P _{1/2}	255.5678	255.4911	1173.0448	0.703	3.000e+07	–	–	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Arrow label: vacuum wavelength | reduced dipole matrix element |(U||er||J)| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).

